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NERVE REGENERATION SYSTEM AND METHOD

Abstract

A stimulation device includes an adaptor component. The adaptor component couples a percutaneous lead to the stimulation device. The stimulation device may apply a stimulation signal to target tissue via the adaptor. A surgeon may place the stimulation device in a container and the adaptor component may be disposed outside of the container. Methods describe prolonged stimulation of target tissue via a stimulation device. The prolonged stimulation may be applied during and after a surgical procedure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. Utility application Ser. No. 18/238,155 entitled “NERVE REGENERATION SYSTEM AND METHOD,” filed Aug. 25, 2023, which is a continuation of U.S. Utility application Ser. No. 17/123,331 entitled “NERVE REGENERATION SYSTEM AND METHOD,” filed on Dec. 16, 2020, which is a continuation of U.S. Utility application Ser. No. 16/623,538 entitled “NERVE REGENERATION SYSTEM AND METHOD,” filed on Dec. 17, 2019, which is a 35 U.S.C. 371 national stage filing and claims priority to International Application No. PCT/US2018/039017 entitled “NERVE REGENERATION SYSTEM AND METHOD,” filed on Jun. 22, 2018, which claims the benefit of U.S. Provisional Patent Application No. 62/523,300, entitled “NERVE REGENERATION SYSTEM AND METHOD,” filed on Jun. 22, 2017, each of which are hereby incorporated by reference.

FIELD OF THE INVENTION

[0002] The invention relates generally to nerve regeneration, and more particularly to systems and methods for accelerating or enhancing regrowth of injured or potentially injured nerves undergoing treatment or surgical repair.

BACKGROUND OF THE INVENTION

[0003] Nerve injuries present clinicians with significant challenges in determining the proper course of treatment to restore impaired motor and or sensory function. Ultimately, the severity of the nerve injury and time post injury have the greatest influence on the treatment plan and potential for success. In many cases surgical intervention may be needed to increase the likelihood that control of muscle function or sensation can be regained. Surgical treatment of nerve injuries typically does not provide immediate restoration of function, as nerve fibers must grow from the point of intervention or repair to the target muscle. Nerve fibers grow at a rate of 1 mm/day, and thus recovery takes a significant amount of time.

[0004] Despite advancements in surgical technique and medical device technology, the rate of growth and organization of fiber growth direction remains a significant factor limiting functional outcomes.

[0005] It may be desirable to provide a method for delivering a period of electrical stimulation as soon as possible, prior to or following repair, preferably while still in the operating room, to improve functional outcome. Furthermore, it may be desirable to be able to initiate a period of stimulation in the operating room and continuing into a post-operative setting without interrupting stimulation. It may alternatively or additionally be desirable to initiate stimulation during surgery or after surgery. Moreover, stimulation may be initiated in an office or an operating room and may continue while the patient moves between rooms. Additionally, it may be desirable to deliver repeated periods of stimulation during recovery, without the need to replace electrode(s) before each application.

SUMMARY OF THE INVENTION

[0006] The following presents a summary of this disclosure to provide a basic understanding of some aspects. This summary is intended to neither identify key or critical elements nor define any limitations of embodiments or claims. Furthermore, this summary may provide a simplified overview of some aspects that may be described in greater detail in other portions of this

disclosure.

[0007] A system for stimulating tissue is described herein. The stimulation system may include a stimulation device comprising a housing, control circuitry disposed within the housing, an operative element coupled with the housing and comprising at least one electrode, a percutaneous lead operatively attachable to the stimulation device, and a container operatively receiving the stimulation device, wherein the container comprises at least one connection port that operatively and electrically couples the stimulation device to the percutaneous lead.

[0008] A method for stimulating tissue is described herein. The method may include performing a subcutaneous operating with a handheld stimulation device, placing a percutaneous lead at a target tissue region, wherein the percutaneous lead is operatively attached to a stimulation device that may be handheld, closing an incision, and applying a stimulation signal to the target tissue region with the percutaneous lead and the stimulation device after the closing of the incision.

[0009] The following description and the drawings disclose various illustrative aspects. Some improvements and novel aspects may be expressly identified, while others may be apparent from the description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings illustrate various systems, apparatuses, devices and methods, in which like reference characters refer to like parts throughout.

[0011] FIG. 1 illustrates a partial cross-sectional view of a stimulation device, in accordance with various disclosed aspects;

[0012] FIG. 2 illustrates a side view of the stimulation device of FIG. 1, in accordance with various disclosed aspects;

[0013] FIG. 3 illustrates a percutaneous electrode, in accordance with various disclosed aspects;

[0014] FIG. 4A illustrates another percutaneous electrode, in accordance with various disclosed aspects;

[0015] FIG. 4B illustrates another percutaneous electrode, in accordance with various disclosed aspects;

[0016] FIG. 5 illustrates an adaptor comprising a percutaneous electrode, in accordance with various disclosed aspects;

[0017] FIG. 6 illustrates an adaptor comprising two connectors, in accordance with various disclosed aspects;

[0018] FIG. 7 illustrates the stimulation device of FIG. 1 coupled with an adaptor, in accordance with various disclosed aspects;

[0019] FIG. 8 illustrates a top view of a container for receiving a stimulation device, in accordance with various disclosed aspects;

[0020] FIG. 9 illustrates a top view of another container for receiving a stimulation device, in accordance with various disclosed aspects;

[0021] FIG. 10 illustrates a back view of the container of FIG. 9, in accordance with various disclosed aspects;

[0022] FIG. 11 illustrates a system comprising the container of FIG. 8, the stimulation device of FIG. 1, and electrode connections, in accordance with various disclosed aspects;

[0023] FIG. 12 illustrates a perspective view of a splitter device, in accordance with various disclosed aspects;

[0024] FIG. 13 illustrates a partial cross-sectional, perspective view of a splitter device, in accordance with various disclosed aspects;

[0025] FIG. 14 illustrates a perspective view of the splitter device of FIG. 12 coupled with

percutaneous leads, in accordance with various disclosed aspects;

[0026] FIG. **15** illustrates a perspective view of another stimulation device, in accordance with various disclosed aspects;

[0027] FIG. **16** illustrates an embodiment of the stimulation device in FIG. **15** without a belt, in accordance with various disclosed aspects;

[0028] FIG. **17** illustrates a back view of the stimulation device of FIG. **16**, in accordance with various disclosed aspects; and

[0029] FIG. **18** illustrates an embodiment of the stimulation device in FIG. **15** with one or more percutaneous leads and a return electrode, in accordance with various disclosed aspects.

[0030] The invention may be embodied in several forms without departing from its spirit or essential characteristics. The scope of the invention is defined in the appended claims, rather than in the specific description preceding them. All embodiments that fall within the meaning and range of equivalency of the claims are therefore intended to be embraced by the claims.

DESCRIPTION OF PREFERRED EMBODIMENTS

[0031] Reference will now be made in detail to exemplary embodiments of the present invention, examples of which are illustrated in the accompanying drawings. It is to be understood that other embodiments may be utilized, and structural and functional changes may be made without departing from the respective scope of the invention. Moreover, features of the various embodiments may be combined or altered without departing from the scope of the invention. As such, the following description is presented by way of illustration only and should not limit in any way the various alternatives and modifications that may be made to the illustrated embodiments and still be within the spirit and scope of the invention.

[0032] As used herein, the words “example” and “exemplary” mean an instance, or illustration. The words “example” or “exemplary” do not indicate a key or preferred aspect or embodiment. The word “or” is intended to be inclusive rather than exclusive, unless context suggests otherwise. As an example, the phrase “A employs B or C,” includes any inclusive permutation (e.g., A employs B; A employs C; or A employs both B and C). As another matter, the articles “a” and “an” are generally intended to mean “one or more” unless context suggests otherwise.

[0033] It is noted that the various embodiments described herein may include other components and/or functionality. It is further noted that while various embodiments refer to a stimulator or stimulation device, various other systems may be utilized in view of embodiments described herein. For example, embodiments may be utilized in a variety of surgical procedures. As such, embodiments may refer to a particular surgical procedure for purposes of explanation. It is noted that aspects of embodiments, however, may be utilized for various other procedures.

[0034] This disclosure generally relates to systems and methods that may improve nerve regeneration or neuroregeneration of tissue via electrical stimulation to increase the speed or amount of nerve growth. The terms “nerve” or “nerve tissue” generally refer to any portion of a nerve including, but not limited to, axons, axon terminals, somas, dendrites, or the like, unless context suggest otherwise. Moreover, aspects disclosed herein may be applicable to nerve tissue throughout a body, whether peripheral nervous tissue or otherwise. Further, while embodiments may reference a surgeon performing a particular action(s), it is noted that other users, automated machines, or the like may perform such actions.

[0035] It is noted that described systems and methods may be utilized in combination with various systems and methods for safeguarding against nerve, muscle, and tendon injury during surgical procedures or confirming the identity and/or location of nerves, muscles, and tendons and evaluating their function or the function of muscles enervated by those nerves. The systems and methods are particularly well suited for assisting in nerve regeneration via a device that may also be utilized by a surgeon in identification of nerves and muscles in order to assure nerve and muscle integrity during medical procedures using medical devices such as stimulation monitors, cutting, drilling, and screwing devices, pilot augers, and fixation devices. Further, the systems and methods

may be utilized in surgery so as to identify the nerve and/or to deliver stimulation for nerve regeneration. This stimulation may be delivered during surgery prior to repair or treatment of the nerve injury and/or following repair or treatment of the nerve injury and/or continue post-surgery. It is noted, however, that various disclosed aspects may be utilized independent of such systems and methods.

[0036] For example, a surgeon may utilize a handheld stimulation device to generate a stimulation signal at sufficiently high levels for the purposes of locating, stimulating, and evaluating nerve or muscle, or both nerve and muscle integrity in numerous medical procedures, including, but not limited to, evaluating proximity to a targeted tissue region, evaluating proximity to a nerve or to identify nerve tissue, evaluating nerve integrity (i.e., following a traumatic or repetitive motion injury) to determine if a repair may be needed, evaluating muscle contraction to determine whether or not the muscle is innervated and/or whether the muscle is intact and/or whether the muscle is severed, identifying specific nerve branches or fascicles for repair or transfer, and evaluating muscle and tendon length and function following a repair or tendon transfer prior to completing a surgical procedure. Before, after or during a procedure, a surgeon may place an electrode or lead on or near the nerve to be stimulated and/or proximal to the site of injury or repair. The electrode may be percutaneous or non-percutaneous (e.g., surface electrode). In an aspect, a percutaneous lead may be taped or otherwise held in place on a patient's skin. This may allow for easy removal after the prolonged stimulation. One exemplary embodiment of such comprises a patch that may be adhered to the skin of a patient. The patch may generally circumscribe the insertion point of the percutaneous lead and may allow a portion of the lead to extend therethrough. In this embodiment, a connector may be utilized to operatively couple of the percutaneous lead with the stimulation device. Alternatively, the lead may be operatively coupled with the patch and the patch may include an adapter that operatively couples with the stimulation device. The patch may include an electrical path between the percutaneous lead and the adapter such that electrical stimulation may pass from the stimulation device through the patch and to the percutaneous lead. The lead may be coupled to a stimulation device that was used during surgery, or another stimulation device, via a wire or other connector. The stimulation device may be placed in a sealed housing to prevent contamination when leaving the operating room into a post-operative setting or to maintain a fixed position. The housing may comprise a port that may allow the lead to be coupled with the stimulation device, and another port to connect a percutaneous or surface electrode, such as a patch, as a return current path. The stimulation device may generate a signal to stimulate the nerve tissue.

[0037] In an aspect, the stimulation device may generate the signal to stimulate the nerve tissue following a procedure or prior to surgical completion. Placement of a percutaneous lead may allow a surgeon to place the lead and stimulate the nerve with a stimulation device during or after the surgery, without requiring the surgeon to hold the stimulation device in place. In an aspect, the stimulation may take place generally after a procedure for a predetermined period (e.g., i minutes, where i is a number). In at least one embodiment, the stimulation may take place for about an hour or less after a procedure has been completed. Stimulation immediately after a procedure (e.g., nerve transfer or nerve release) may increase the speed, quality, or amount of nerve regeneration. An aspect enables the onset of stimulation to begin in the operating room, prior to procedure completion, and continuing into a post-operative setting, without disrupting stimulation. Reduction of the delay from the completion of surgical intervention to start of stimulation, may provide further clinical benefit.

[0038] In one example, electrical stimulation may be applied to determine a baseline level of nerve excitability at the start of a surgery, i.e., a threshold test. During or near completion of the surgery a second test of nerve excitability may be conducted. This second test may be compared against the first. If the second test results in lower nerve excitability, prolonged stimulation may be applied after the surgery to help with nerve regeneration. This entire series of tests may be done with a single electrode and stimulation device or may be accomplished with two or more electrodes or

stimulation devices. Further still, the initial threshold test may be done with a different stimulation and the same lead or a different stimulation device and lead or with the same stimulation device and the same lead. The application of the second test may determine if the nerve regeneration stimulation is necessary or desirable for the patient, i.e., to treat any potential nerve injury or trauma as suggested by the reduction in nerve excitability between the first and second test. The threshold test may comprise electrical stimulation to a nerve or nerves within a target tissue region to determine or measure the excitability of the nerve or nerves. In such situations, it may only be necessary to test nerve excitability to determine if the nerve regeneration therapy is needed prior to applying such nerve regeneration therapy. Threshold testing may be done at any time during the surgery, such as when a retractor is removed, a limb of a patient has a force applied to it or the like. [0039] In another example, stimulation of a nerve may be applied during surgery prior to treatment or repair of nerve damage, suspected nerve damage, a risk of future nerve damage, or the like. A percutaneous lead may be placed at or near a nerve (e.g., within a range of the signal) to allow a stimulation device to generate and apply a signal to the nerve tissue. The stimulation may take place for a predetermined period of time and may be conducted at any time during surgery. For example, stimulation may occur immediately at the start of surgery and may be conducted for an hour or less. It is noted that the stimulation may be apply at different strengths, frequencies, patterns, or the like. Stimulation at time of a surgical operation may increase the speed, quality, or amount of nerve regeneration or nerve function recovery. The percutaneous lead may be anchored, held, or otherwise left in place after stimulation, during, and/or after a surgical operation.

[0040] As described herein, the parameters of the stimulation signal of the prolonged stimulation may be preprogrammed or may be set by a surgeon. In at least one embodiment, the pulse width may be held constant (e.g., not adjusted) during the stimulation. In an aspect the stimulation signal may be applied at generally between 10-100 Hz, 15-30 Hz, or about 16 Hz. The amplitude of the stimulation may be held constant or be adjusted by the surgeon. In general the stimulation amplitude will generally be between 0.1-20 mA, typically between 1.0-2.0 mA. Moreover, the stimulation signal may be applied for a prolonged period (e.g., an hour). It is noted that the prolonged stimulation signal may be applied in a single dose or multiple doses, or for durations up to or more than 8 hours. For example, multiple doses may be applied through the same percutaneous lead, which may be kept in place between doses or may be removed between the doses. In such examples, one lead may be utilized and a different stimulation device utilized depending upon the location of the patient during stimulation, i.e., surgical location or post-operative location. In the alternative, one lead may be utilized and a single stimulation device utilized regardless of the location of the patient during stimulation. In yet another alternative, a different lead and different stimulation device may be utilized depending upon the location of the patient during stimulation. In such dosing examples, the electrical stimulation may be applied for a period of time, such as k minutes, where k is a number (e.g., 1, 2, 5, 10, 15, 60, etc.). In an example, the period may be between about 10 minutes and an hour or more.

[0041] For example, stimulation systems described herein may intraoperatively deliver, for a period of time, an electrical stimulation to repaired peripheral nerves. An attachment (e.g., a percutaneous electrode lead attachment such as those described in U.S. Patent Application US2014/0073985A, which is incorporated by reference herein) may be coupled with a stimulation device after a surgical procedure. It is noted that the stimulation device may be a stimulation device that was used during the procedure. For instance, a surgeon may utilize a handheld stimulation device to access nerve function during a surgical operation. The surgeon may place one or more implantable leads at a position where the lead may stimulate nerves that may have damage, potential for damage, or may require recovery after completion of the surgical operation. It is important to note that the present disclosure contemplates providing the electrical stimulation to be preventative, i.e., to mitigate the impact of a nerve injury, risk of nerve injury, or suspect nerve injury. The surgeon may, for instance, place the handheld stimulation device into a sterilized container and may electrically

attach the one or more implantable leads to the handheld stimulation device via ports of the container. As such, the patient's nerve may be stimulated post operatively via the same stimulation device utilized for the surgical operation. Once post-operative stimulation is completed (e.g., after 1 hour of stimulation), the stimulation device may be discarded or otherwise removed.

[0042] The stimulation device may be disposable or reusable. Described systems may allow a stimulation electrode to be placed intraoperatively, in close proximity to the nerve to be stimulated; allowing the lead to pass out of the tissue and attach to the stimulation device.

[0043] The stimulation device may provide a prolonged stimulation after an incision is closed. At a desired stopping time, the lead may be removed post operatively, and the prolonged course of stimulation may be discontinued. In an aspect, this may eliminate the need for extended operating time, and may free up a surgeon or staff from holding the stimulator on the nerve for prolonged stimulation. The lead may also be left in for a period of time, to enable repeated doses of stimulation across multiple days or weeks without need for placement of another lead.

[0044] The surgeon may place the stimulation device in a container. The container may comprise a bag, box, or other container. The container may seal the stimulation device within the container to prevent or reduce the chance of contamination, provide stability, or the like. In at least one embodiment, the container may comprise one or more ports that may allow the stimulation device to be coupled with a wire or lead. The return current electrode may be attached to a side of the container that is in contact with a patient, or may be an external electrode that connects to a second port on the container. As described herein, the container may comprise a suitable material such as a plastic, vinyl, metal, or other material.

[0045] It is noted that the disclosed systems and methods are applicable for use in a wide variety of medical procedures involving peripheral nerves. By way of non-limiting example, the various aspects of the invention have application in treatment of nerve transection injuries, nerve crush injuries, suspected nerve injuries, risk of nerve injuries or function reduction, or nerve transfer procedures, including, without limitation nerve decompression procedures (such as carpal tunnel or cubital tunnel syndrome), neurolysis, nerve transfer, nerve repair (such as direct repair, autograft, allograft, or conduit), and iatrogenic injury (thermal, stretch, compression, or transection).

[0046] In at least one embodiment, an electrical stimulation system may comprise a stimulation device and an adaptor. The stimulation device may comprise housing, control circuitry operatively generating a stimulation signal, wherein the control circuitry is disposed within the housing, and an operative element coupled with the housing and comprising at least one electrode. The adaptor may be selectively attached to the operative element, and may comprise a percutaneous lead electrically coupled to the stimulation device through the at least one electrode, the percutaneous lead insertable into a patient during a subcutaneous surgery and after the subcutaneous surgery and wherein the stimulation device is capable of applying electrical stimulation during the subcutaneous surgery and after the subcutaneous surgery. The percutaneous lead may comprise a twisted wire.

[0047] The control circuitry may apply the electrical stimulation after the subcutaneous surgery for nerve regeneration therapy. Nerve regeneration therapy may comprise stimulation to a nerve to alter recovery (e.g., improve, enhance, accelerate, etc.) of the nerve so stimulated. In another aspect, the control circuitry may apply the electrical stimulation for a period between **10** minutes and one hour. The electrical stimulation system may further comprise a container operatively receiving the stimulation device, wherein the container comprises at least one connection port that operatively and electrically couples the stimulation device to the percutaneous lead. In at least one embodiment, the electrical stimulation system may further comprise a splitter device, operatively coupled to the stimulation device, electrical stimulation system, and at least one other lead, wherein the splitter device receives the stimulation signal and generates generally uniform output signals to the lead and the at least one other lead. The splitter device may be disposed within the container. The container may comprise an attachment device that operatively attaches the container to an

object. The container may additionally or alternatively comprise one or more fasteners to selectively secure the control circuitry within the container. It is noted that the at least one connection port may comprise a return port that operatively receives a return electrode of the stimulation device from within the container. The return port may operatively and electrically couple the return electrode to a return lead.

[0048] In another aspect, embodiments include a method for stimulating tissue may comprise In another aspect, embodiments include a method for stimulating tissue may comprise performing a subcutaneous surgery with a stimulation device, placing a lead within range of a target tissue region, wherein the lead is operatively attachable to the stimulation device, and applying a stimulation signal to the target tissue region with the lead and the stimulation device before or after the subcutaneous surgery. Placing the lead may comprise placing the lead percutaneously. The stimulation device may be a handheld stimulation device. The method may include placing the stimulation device within a container after performing the subcutaneous surgery and for nerve regeneration therapy. It is noted that the method may include attaching the lead to an external end of a port of the container and attaching the stimulation device to an internal end of the port.

According to at least one example, the stimulation device is placed within the container while in an operating room. Moreover, stimulating the target tissue region with the lead and the stimulation device may be done prior to performing certain portions of the subcutaneous surgery or during the subcutaneous surgery to determine a threshold for excitability of a nerve within the target tissue region, for the purpose of determining if application of prolonged stimulation for nerve regeneration therapy is appropriate. In an aspect, the determined threshold for excitability may be utilized to determine if application of stimulation for nerve regeneration therapy is appropriate.

[0049] A method for stimulating tissue may comprise placing a lead within range of a target tissue region and applying a stimulation signal with a stimulation device; and performing a subcutaneous surgery with the handheld stimulation device before or after applying the stimulation signal. The method may further comprise storing the stimulation device within a container prior to moving out of an operative setting in order to maintain sterility after and stability while moving a patient to another location. In at least some embodiments, the method may comprise stimulating the target tissue after performance of the subcutaneous surgery.

[0050] Turning now to FIGS. 1 and 2, there is a stimulation system **100** that may comprise a stimulation device **102** configured for locating, monitoring, and stimulating tissue and other structures throughout the body. The stimulation system **100** may be utilized for locating and identifying tissue and safeguarding against tissue and/or bone injury during surgical procedures. In another aspect, the stimulation device **102** may be utilized for percutaneously stimulating a nerve after a surgery for a desired amount of time. In an aspect, the stimulation may be post-operative stimulation after a surgical procedure.

[0051] The stimulation device **102** may include or be coupled with one or more attachments or operative elements including, for example, a probe **110** (e.g., which may be blunt, needle-like, etc.), a cutting device, a drilling or screwing device, a pilot auger, and a fixation device. It is noted that attachments may be removable, attachable, or permanently affixed to the stimulation device **102**. It is noted that while embodiments may describe use of a particular attachment (e.g., probe **110**) for simplicity of explanation, the various embodiments may utilize other types of attachments.

[0052] In an exemplary embodiment, stimulation device **102** comprises control circuitry **104**, disposed in a housing **120**, that may apply a stimulation signal to a desired tissue region. The control circuitry **104** may be coupled to a power source, such as a battery, power mains, or the like. The control circuitry **104** may generate the stimulation signal with desired parameters, as described herein. In an aspect, a user may adjust parameters and/or control the control circuitry **104** to generate the stimulation signal via one or more user interfaces **108**, which may comprise at least one of a switch, button, slide, touch screen, or the like.

[0053] For instance, a user may grasp the stimulation device **102** via the housing **120**. The housing

120 may include gripping portion **122**. The gripping portion **122** may comprise indents, protrusions, elastomeric material, roughened material or other features that may aid in the user grasping the stimulation device. The gripping portion **122** of the housing **120** may include an over molded portion that may comprise all or part of the length of the housing **120**. In an aspect, the over molded portion may comprise a thermoplastic elastomer material. It is noted that gripping portion **122** may be removable, attached to, or integrally formed with the housing **120**.

[0054] In an example, a user may position the probe **110** so that an uninsulated or stimulating portion or electrode **114** is at a desired location. The user may interact with one or more of the user interfaces **108** to control delivery of stimulation signal, generated by the control circuitry **104**, to the desired tissue region. The gripping portion **122** may aid in a user's efforts to hold the stimulation device **102**. In an aspect, the control circuitry **104** communicates the stimulation signal to the stimulation probe **110** via a lead **112** that may travel through an insulated portion **112** of the probe to the uninsulated portion **114**.

[0055] It is noted that the probe **110** may comprise one or more flexible materials (e.g., metal, plastic, etc.) so that a user may bend or otherwise manipulate the probe **110**. In another aspect, the stimulation device **102** may comprise a nose cone **124** that may be flexible or rigid. An operative element (e.g., probe **110**) may extend from the proximal end of the nose cone **124**. The user may apply pressure to the nose cone **124** so that it moves or otherwise manipulates the probe **110**. This may allow a surgeon or other user to position the uninsulated portion **114** at a desired position of a targeted tissue region. For example, the uninsulated portion **114** of the probe **110** is positioned in electrical conductive contact with at least one of muscle, nerve, or other tissue.

[0056] A flexible nose cone **124** may allow the surgeon to use either a finger or a thumb positioned on the nose cone **124** to make fine adjustments to the position of stim probe **110** at the targeted tissue region. The surgeon may grasp the housing **120** with the fingers and palm of the hand, and position the thumb on the nose cone **124**, and with pressure applied with the thumb, cause the probe **110** to move while maintaining a steady position of the housing **120**. This flexible nose cone **110** may allow for increased control of the position of the probe **110** with the movement of the surgeon's thumb (or finger, depending on how the stimulating probe is held). In another aspect, the nose cone **124** may comprise gripping components, such as ribs, indents, roughened surfaces, or the like.

[0057] It is noted that the nose cone **124** may comprise a single piece or it may comprise one or more pieces attached together. For example, nose cone **124** may comprise an inner portion that may include thermoplastic material having flexibility (e.g., LUSTRAN® ABS 348, or similar material), and an outer portion that may comprise a softer over molded portion and may be made of a thermoplastic elastomer material having flexibility (e.g., VERSAFLEX™ OM 3060-1 from GLS Corp). It is noted, however, that nose cone **124** may be generally rigid in at least some embodiments.

[0058] While described as a “cone” nose cone **124** may comprise a generally tapered shape. It is noted, however, that nose cone **124** may comprise other or different shapes (e.g., rounded, squared, prism, conical, etc.). Moreover, in embodiments, stimulation device **102** may not comprise a nose cone **124**, such that probe **110** extends directly from housing **120**.

[0059] As described herein, a stimulation signal may flow from the stimulation device **102** through the lead **112** to the probe **110**, which may act as an electrode. The stimulation system **100** may include one or more other electrodes, such as a return electrode, as described herein. For instance, in monopolar operation, a return electrode (or indifferent electrode) provides a return path for electrical signals passing through the tissue, and returning to the stimulation device **102**. It is noted that stimulation system **100** may operate in a monopolar, bipolar or other configurations, as described here as well as elsewhere in this disclosure.

[0060] In various embodiments, the control circuitry **104** may generate the stimulation signal to operatively generate a physical motor response of a tissue (e.g., muscle, innervated muscle, nerve,

etc.). The physical motor response may indicate whether the stimulation signal was delivered and/or whether a sufficient stimulation signal was delivered. For example, the motor response may include a physical motor response (e.g., twitching or contraction).

[0061] In another aspect, the stimulation device **102** may generate one or more visual or audio signals (e.g., via a speaker (not shown)), which indicate to the surgeon the status or diagnostic information. For instance, stimulation device **102** may comprise an indicator light **126**. The indicator light may comprise one or more light sources, such as a light emitting diode (LED). In an aspect, the indicator light **126** may comprise a translucent (e.g., semi-translucent, fully translucent, etc.) surface that operatively shines or disperses light from an internal light source (not shown). In an aspect, the light source may generate light in one or more colors (e.g., green, yellow, blue, red, etc.), patterns (e.g., blink rate, pattern of colors, etc.), or the like. According to embodiments, the status or diagnostic information may indicate whether the stimulation signal was delivered and/or whether a sufficient stimulation signal was delivered. For example, the status or diagnostic information may indicate that an electric signal was returned from tissue, which may indicate sufficient proximity, contact, or delivery of a stimulation signal via an operative element (e.g., probe **110**). In another aspect, the indicator light **126** may indicate that the stimulation device **102** is on/off, producing or not producing a stimulation signal, or the like.

[0062] In an example, the indicator light **126** allows the surgeon to confirm delivery of stimulus current to tissue. Through the use of different tones, colors, different flash rates, etc., the indicator **126** allows the surgeon to confirm that the uninsulated tip **114** is in place, the instrument is turned ON, and that stimulus current is flowing with sufficient delivery to tissue. Thus, the surgeon has a much greater confidence that the desired stimulation amplitude is being delivered to the nerve, as in the case of a nerve transfer or nerve graft, a muscle contraction will not be observed since the nerve is no longer in continuity. These indicators can be checked periodically to ensure stimulation is being delivered for the desired duration (e.g. between about 1 minutes and one hour).

[0063] As another example, in use the indicator **126** may be configured to illuminate continuously in one color when the stimulation device **102** is turned on but not in contact with tissue. After contact with tissue is made, the indicator **126** may flash (i.e., blink) to indicate that stimulation is being delivered. If the stimulation has been requested, i.e., the stimulation probe has been turned on, but there is no stimulation being delivered because of a lack of continuity between the probe **110** and the return electrode **130**, or an inadequate connection of the probe **110** or the return electrode **130** to the patient tissue, the indicator **126** may illuminate in a different color, and may illuminate continuously or may flash.

[0064] As described herein, the indicator **126** may comprise a ring that provides a visual indication around at least a portion, and desirably all of the circumference of the stimulation device **102** generally near the nose cone **124**. A ring indicator may be an element of the gripping portion **122**, or it may be an element of the flexible nose cone **124**, or the ring indicator may be positioned between the gripping portion **122** and the nose cone **124**. The ring may also include a reflective element to improve and focus the illumination effect of the light emitting source, e.g., one or more LEDs. The ring and the reflective element may be a single component, or more than one component. Audio feedback also makes possible the feature of assisting the surgeon with monitoring nerve integrity during surgery.

[0065] While stimulation device **102** is described as generating an indication, it is noted that various other components of the stimulation system **100** may generate all or part of the indication. For instance, the stimulation device **102** (or a separate device) may monitor delivery of the stimulation signal. The stimulation device **102** may transmit status and diagnostic information (e.g. delivered current, stimulation duration, contraction presence, or the like) to a separate device (e.g., laptop, wearable electronic device, cellular phone, tablet, computer, speakers, light source, or the like). In an aspect, the stimulation device **102** may include a communication component that may be wired or wireless. For example, the stimulation device **102** may include a wireless

transmitter/receiver configured to communicate via one or more communication protocols (e.g., Wi-Fi, BLUETOOTH, NFC, etc.).

[0066] In embodiments, stimulation device **120** may comprise a hand-held stimulation device. Housing **120** may be generally tubular, hexagonal, or other elongated shape. According to an aspect, the housing **120** may be ergonomic and sterile for use in operative procedures. For instance, the stimulation device **120** may be packaged in a sealed container that may allow a surgeon to open and use the stimulation device **120** without the need for sterilization. It is noted, however, that parts of the stimulation system **100** may be sterilized, such as probe **110**. In another aspect, the stimulation device **120** may comprise a single use instrument for use during surgical procedures to identify nerves and muscles, muscle attachments, contract muscles to assess the quality of surgical interventions or the need for surgical interventions, evaluate the function of nerves already identified through visual means, or provide prolonged stimulation of a nerve.

[0067] The stimulation device **120** may be sized small enough to be held and used by one hand during surgical procedures, and may be ergonomically designed for use in either the left or right hand. In an embodiment, the stimulation device **120** may have a width of about 20 millimeters to about 30 millimeters, and desirably about 25 millimeters. The length of the stimulation device **120** (not including an operative element) may be about 18 centimeters to about 22 centimeters, and desirably about 20 centimeters. An operative element (e.g., probe **110**) may also include an angle or bend **116** to facilitate assess to deep as well as superficial structures without the need for a large incision. As illustrated, the bend **116** may be generally downward, relative the directions shown in FIGS. **1** and **2**. In an aspect, this may allow a surgeon to maintain a line of sight with target tissue and/or the uninsulated portion **114**.

[0068] In one or more embodiments, as described here as well as elsewhere in this disclosure, an operative element may be mono-polar or bi-polar. For instance, probe **110** may be mono-polar. A return electrode **130** may be coupled to control circuit **104** via an insulated wire **132**. The return electrode **130** may comprise any of a variety of electrode types (e.g., paddle, needle, wire, or surface electrode). In another aspect, the stimulation device **102** may be bipolar and may comprise a return electrode in the probe **110** or other operative element.

[0069] User interfaces **108** may allow a user to turn ON/OFF the stimulation device **102** (or set to standby), and may allow a user to control the stimulation signal amplitude selection within a predefined range (e.g., 0.1 0.5, 2.0, and/or 20 mA). In configurations, user interface **108** may be a four or five position switch. It is noted that the user interface **108** may allow for selection and change of frequencies within a range. Before the first use of the stimulation device **102**, the user interface **108** is in the OFF position and keeps the stimulation probe off. After the user interface **108** has been turned ON (e.g., by moving the switch **155** to an amplitude selection), the OFF position now corresponds to a standby condition, where no stimulation would be delivered. In one embodiment, once the stimulation device has been turned on, it cannot be turned off, it can only be returned to the standby condition and will remain operational for a predetermined time, e.g., at least about seven hours. This may allow the stimulation device **102** to be only a single use device, so it cannot be turned OFF and then used again at a later date. It is noted, however, that some embodiments may allow the user to turn off the stimulation device **102** after it has been turned on. In one example, the user interfaces **108** may allow for selection of “prolonged stimulation.” Once prolonged stimulation has been selected, the stimulation device **102** may disable user control of certain stimulation parameters, may allow the stimulation device **102** to be turned off, or may turn off after a certain time in the prolonged stimulation mode (e.g. 1 hour).

[0070] The user interfaces **108** may allow for adjustment of a stimulation signal pulse width from a predefined range (e.g., about zero to about 200 microseconds). In one embodiment, the user interfaces **108** may be a potentiometer to allow a slide control to increase or decrease the stimulation signal pulse width within the predefined range. The stimulation pulse may have a non-adjustable frequency in the range of about 10 Hz to about 30 Hz, and desirably about 16 Hz. In

some embodiments, the stimulation pulse may comprise an adjustable frequency.

[0071] As a representative example, the stimulation pulse may have a biphasic waveform with controlled current during the cathodic (leading) phase, and net DC current less than 10 microamps, switch adjustable from about 0.1 milliamps to about 20 milliamps, and pulse durations adjustable from about zero microseconds up to about 1 millisecond.

[0072] The operative element (e.g., probe **110**) exits or attaches to the housing **120** at the nose cone **124** to deliver stimulus current to the excitable tissue. The probe **110** comprises a length and a diameter of a conductive material, and is desirably fully insulated with the exception of the uninsulated portion **114**, e.g. about 1.0 millimeters to about 10 millimeters, and desirably about 4 millimeters to about 6 millimeters, which is non-insulated and serves as the stimulating to allow the surgeon to deliver the stimulus

[0073] The size of the uninsulated portion **114**, (the active electrode) of the probe **110** ensures a high current density that will stimulate nearby excitable tissue. The insulation portion **112** may comprise a medical grade heat shrink.

[0074] The conductive material of the probe **110** comprises a diameter having a range between about 0.5 millimeters to about 1.5 millimeters, and may be desirably about 1.0 millimeters. The length of the operative element **110** may be about 50 millimeters to about 60 millimeters, although it is to be appreciated that the length may vary depending on the particular application. As shown, the probe **110** may include one or more bends to facilitate accurate placement of the uninsulated portion **114**. In one embodiment, the conductive material of probe **110** is made of a stainless steel, solid wire, although other conductive materials may be used. Further, the probe **110** may include an anchor **116**. The anchor **116** may be of any appropriate configuration. By way of a non-limiting example, the anchor **116** may comprise a bend in the probe **110** such that upon insertion of the probe **110** into tissue of a patient, or more specifically, the anchor **116** being inserted into tissue of the patient, the anchor **116** prevents an undesired withdrawal of the anchor **116** and/or probe **110**.

[0075] As previously described, in monopolar operation, a return electrode **131** (or indifferent electrode), for example, provides an electrical path from the body to the stimulation device **102**. The return electrode **130** may be placed on the surface of intact skin (e.g., surface electrodes as used for electrocardiographic or electromyographic monitoring during surgical procedures) or it might be needle-like and be placed in the surgical field or penetrate through intact skin or an incision.

[0076] The configuration of the stimulating medical devices that form a part of the system can vary in form and function. Various representative embodiments of illustrative medical devices will be described.

[0077] Referring now to FIGS. 3-4A, there are percutaneous electrodes **300** and **400** in accordance with various disclosed aspects. Percutaneous electrodes **300** and **400** may generally include an insulated body **302/402**, an anchor **304/404**, and one or more uninsulated portions **306/406**. It is noted that percutaneous electrode **300** may comprise similar aspects as percutaneous electrode **400**, unless context suggest otherwise or specific reference is made to a difference between the two. As such, while examples may refer to one of the percutaneous electrodes **300** and **400** for simplicity of explanation, the other may be utilized. Moreover, various other percutaneous electrodes may be utilized by embodiments disclosed herein.

[0078] In an example, percutaneous electrode **300** may be placed at or near a target tissue region and may be coupled with a percutaneous lead or wire, as described herein. It is noted that percutaneous electrode **300** may be positioned while an incision is open and may be left in place while the incision is closed. In at least one other embodiment, percutaneous electrode **300** may be positioned when an incision is closed or by deploying the percutaneous electrode **300**.

[0079] In embodiments, percutaneous electrode **300** may comprise strands of stainless steel wire insulated with a biocompatible polymer. Each wire strand may have a diameter of approximately **34** μm and the insulated multi-strand lead wire may have a diameter of approximately **250** μm . It should be understood, however, that these dimensions are merely exemplary and the present

teachings are not limited to such. Any appropriate sized, shaped and configured electrode and percutaneous lead may be used. The insulated wire may be formed into a spiral or helix as has been found to accommodate high dynamic stress upon muscle flexion and extension, while simultaneously retaining low susceptibility to fatigue. The outer diameter of the percutaneous electrode **300** may be approximately 580 μm and it may be encased or filled with silicone or the like. In at least some embodiments, percutaneous electrode **300** may be made out of a different material (e.g., another metal, conducting polymer), may be insulated with another material, or may not be insulated. Further, the lead may be cylindrical or paddle-like.

[0080] Unlike surface electrodes that are applied to the surface of the patient's skin using an adhesive, percutaneous electrode **300** may be surgically implanted or otherwise inserted into select tissue. The terminal end or anchor **304** may comprise one or more tines **308** (e.g., tines **408** of percutaneous electrode **400**). The anchor **304** may be insulated or uninsulated. In at least some embodiments, the anchor **304** may be inserted directly into tissue and may deliver stimulation signals to the tissue. In another aspect, the anchor **304** may generally hold the percutaneous electrode **300** in place. For instance, the one or more tines **308** may comprise a bend, curve, barb, etc., that prevents the percutaneous electrode **300** from substantially moving or unintentionally coming loose. As shown in FIGS. **3** and **4**, disclosed embodiments may include different types of anchors **304/404**. For instance, an anchor may include j tines, where j is a number. In an exemplary embodiment, a patch assembly may be utilized in conjunction with the percutaneous electrode **300**. The patch assembly may comprise several layers, including an adhesive layer, an electrode layer, a reinforcement layer and a cover layer. In one embodiment, the patch assembly may include a power source for the stimulation device. Further, the patch assembly may act as a surface electrode. In one embodiment, the patch assembly may include engagement member or members that electrically couple the stimulation device to the percutaneous electrode **300** to provide stimulation for nerve regeneration. The engagement member may comprise a snap, a magnetic male and female member capable of operable engagement, a bayonet engagement device, or any known engagement mechanisms capable of electrically coupling the stimulation device with the percutaneous electrode **300**. The present disclosure contemplates any such configuration of the patch assembly.

[0081] In another aspect, an anchor may include threaded members (e.g., screws) or the like. Further still, the percutaneous electrode **300** may not include any tines or anchors. In these embodiments, the percutaneous electrode **300** may be placed near or around, i.e., generally circumscribing all of or a portion of the applicable nerve. Further, the percutaneous electrode **300** may be placed over, i.e., on top of or at the bottom of, the applicable nerve, or near, i.e., in an operative distance from the applicable nerve in any manner. The present teachings are not limited to a specific configuration. Embodiments may include a nerve cuff, a coiled lead, a straight lead, lead with a hook, lead with a tine or tines, or the like.

[0082] According to embodiments, percutaneous electrode **300** may comprise flexible materials that allow some or all of the percutaneous electrode **300** to bend or deform. In an example, the insulated portion **302** may be a lead that is generally flexible to allow removal, positioning, or other manipulation of the percutaneous electrode **300**.

[0083] In embodiments, sections of the uninsulated portion **306** may be separated by insulated portions **310** (e.g., insulation portions **410** in FIG. **4A**). It is noted that different sections of the uninsulated portions **306** may be electrically isolated from each other to allow for bipolar stimulation. In another aspect, the insulated portions **310** may allow for increased strength, positioning, or the like of the percutaneous electrode **300**. The uninsulated portion **306** may operatively deliver a stimulation current. It is noted that the uninsulated portion **306** may be disposed anywhere along the percutaneous electrode **300**. For instance, the uninsulated portion **306** may be disposed at one or more tines **308**, at anchor **304**, or the like.

[0084] FIG. **4B** illustrates another percutaneous electrode **450** comprising an insulated body **452** and an anchor **454**. The anchor **454** may comprise a twisted or braided wire. The anchor **454** may

be electrically conducting and not insulated such that it may apply a stimulation signal to tissue. In an aspect, the anchor **454** may comprise a helical portion **458**. In another aspect, body **452** may include twisted, braided or helical portion **460**. The helical portion **458** and helical portion **460** may anchor the percutaneous electrode **450** in places. In an aspect, the percutaneous electrode **450** may allow for extended use or implantation. For example, tissue may heal around the helical portion **458** or helical portion **460**. The shape of these portions allows tissue to grasp and grow in between turns or bends of the helical portion **458** and helical portion **460**. The tissue growth will anchor the percutaneous electrode **450** and may prevent or reduce chances of developing infections as the tissue heals around the percutaneous electrode **450**. This may allow the electrode **450** to remain in place for an extended period of time or for a short period of time. Further, the helical portion **460** may comprise a fine-coiled wire with a insulative material surrounding such.

[0085] Turning now to FIGS. 5-7, with reference to FIGS. 1-4, there are adaptor **500** (which includes percutaneous electrode **400** (and may also include electrode **300** and **450**) and a lead wire **502**), adaptor **600** (which may be coupled to a stimulation device and/or percutaneous electrode), and stimulation system **700** (which may include stimulation device **102**). It is noted that liked named components of the various embodiments may comprise similar aspects, unless context suggests otherwise or a particular distinction is made.

[0086] In an embodiment, adaptor **500** may primarily include percutaneous electrode **400**, wire **502** and connector **512**. Wire **502** may connect terminal end **412** of the percutaneous electrode **400** with connector **512**. In an aspect, wire **502** may comprise an insulated wire that is removably or irremovably attached to the percutaneous electrode **400** and/or connector **512**. As described herein, the adaptor **500** may be configured to allow a stimulation probe **110** to deliver a stimulation signal below the skin of a subject patient.

[0087] In an aspect, connector **512** may include an opening **514** that may receive an operative element. As shown in FIG. 7, the opening **514** may receive the probe **110** of a stimulation device **102**. The opening **514** may be tapered to maintain the probe **110** in a friction fit within the connector **512**. The connector may further include other retaining features, such as a fastener (e.g., screw, clasp, threaded portions, VELCRO, magnet, etc.) to retain the connection between the connector **512** and the probe **110**. It is noted that connector **512** may comprise an electrical connection disposed within the connector **512** that may operatively couple an uninsulated or stimulating portion of the probe **110** with the wire **502**.

[0088] Wire **502** may extend from the connector **512**. Wire **502** may be an electrical conductor in electrical connection with probe **110** when probe **110** is operatively inserted into the connector **512**. It is noted that the wire **502** may be any appropriate length, such as 24 inches or a length between 12 inches and 48 inches. The lead wire may further be any appropriate gauge, such as 24 AWG wire.

[0089] Percutaneous electrode **400** may be coupled to the wire **502** at a terminal end **412**. According to an embodiment, the wire **502** may be removably or irremovably coupled to the terminal end. It is noted that the wire **502** may be coupled directly to the terminal end **412** and/or may be coupled indirectly to the terminal end **412**, such as through one or more other connectors (not shown). Moreover, wire **502** may be coupled to other portions of the percutaneous electrode **400**. In an aspect, the connection between the wire **502** and the percutaneous electrode **400** may be insulated or uninsulated.

[0090] As shown in FIG. 6, adaptor **600** may comprise a wire **602** and one or more connectors **612/622**. Each connector **612/622** may comprise an opening **614/624**. In an aspect, openings **614/624** may comprise similar or different dimensions. For instance, openings **614/624** may be operatively sized and shaped to receive an operative element and/or a percutaneous electrode (e.g., percutaneous electrode **300/400**). In at least one embodiment, opening **614** is operatively sized to receive an operative element, and opening **624** is operatively sized to receive a percutaneous electrode. In another aspect, connectors **612/622** may comprise elastomeric materials that may

stretch, compress, or otherwise fit different sized components. Moreover, while embodiments disclose connectors **612/622** as female connectors, it is noted that one or more of connectors **612/622** may be a male connector. In another aspect, wire **602** (or **502**) may comprise one or more branches or pathways such that connector **614**, for example, may be electrically coupled with one or more other connectors through wire **602**.

[0091] Turning to FIG. 7, and as described herein, system **700** may comprise stimulation device **102** that may be coupled with an adaptor **500** (or other described adaptors). In an aspect, a surgeon may utilize stimulation device **102** during a procedure (e.g., location of a nerve, nerve assessment, etc.) When the procedure is complete, the surgeon may place a percutaneous electrode in a desired location. The surgeon may couple the electrode to the stimulation device **102** via connector **500**.

[0092] As illustrated, connector **512** may be attached to the probe **110**. The surgeon may utilize user interfaces **108** to select a stimulation process. For instance, the surgeon may operatively set the stimulation device **102** to deliver a prolonged stimulation to target tissue. In an aspect, prolonged stimulation may follow completion of a surgical intervention (e.g. nerve repair, nerve release, or nerve transfer). In an aspect, prolonged stimulation may be delivered to a nerve or muscle, distal to site of surgical intervention, to increase muscle viability while the nerve re-grows.

[0093] It is noted that the stimulation device **102** may comprise a preprogrammed stimulation process that may operatively generate stimulation signals for prolonged stimulation. In another aspect, user interfaces **108** may allow a user to manually program or adjust stimulation parameters, such as intensity, pattern, time, or the like.

[0094] FIGS. **8-10** illustrate a container **800** that may operatively receive a stimulation device (e.g., stimulation device **102**). FIG. **11** illustrates the stimulation device **102** disposed within the container **800** and coupled with adaptor **500**. The container **800** may operatively hold the stimulation device for transitioning a patient to post-op or during post-op. This container may be used to hold a stimulation device that was previously used during the procedure, isolating the device with potential biological contamination (e.g. blood) from the rest of the environment. It is noted that container **800** may comprise various materials, such as one or more of plastic, metal, or the like. In at least one embodiment, container **800** may comprise a sterilized material.

[0095] A body **802** of the container **800** may comprise a terminal end **804** and a proximal end **806**. The terminal end **804** may comprise an opening **808** that may allow a surgeon to insert the stimulation device **102** in the container **800**, and seal the container. It is noted that various other sides or portion of the body **802** may comprise an opening to allow the surgeon to place the stimulation device **102** in the container **800**.

[0096] According to an embodiment, container **800** may comprise one or more fasteners **810** that may fasten or hold the stimulation device **110** when inserted into the container **800**. The fastener **810** may include a hook, clasp, screw, adhesive, or other fastener. For example, fastener **810** may comprise a clasp that may be sized and shaped to allow body **120** of stimulation device **102** to snap into the clasp. In an aspect, the clasp may friction fit with the body **120** to maintain the stimulation device **102** in a general position relative the container **800**.

[0097] Body **802** may comprise one or more ports, such as port **812** and port **814**. Port **812** and port **814** may be disposed at various locations, in an example, port **812** is disposed at a location to allow an operative element of stimulation device **102** to be inserted or otherwise coupled to port **812** when the stimulation device **102** is attached to the fastener **810**. Port **814** may operatively receive return electrode **130**. It is noted that port **812** and port **814** may be positioned generally proximal each other or may be positioned at other locations.

[0098] It is noted that ports **812** and **814** may allow components disposed within the container to be electrically coupled to components disposed outside of the container. In an example, the port **812** may receive a connector (e.g., connector **512** as shown in expanded view **1100**). For instance, port **812** may receive a portion of connector **512** and may operatively hold the connector **512** in place (e.g., via a friction fit, fastener, or the like).

[0099] In at least one embodiment, the port **812** (and/or **814**) may comprise a connector **512** that is built into the container **800**. For instance, a user may place the stimulation device **110** in the container **800** and may insert the port **812** into the connector **512** and the return electrode in another connector of port **814**. The user may seal the container **800**. The user may connect the adaptor **500** to the port **812** from outside of the container **800**. In another aspect, the user may connect a return electrode **1102** to the port **814** via a wire **1104** and/or another connector. According to an aspect, this may reduce or prevent the spread of material (e.g., bodily fluids, etc.) on the stimulation device **110** from exiting the container **800** during post-op procedures.

[0100] In at least one aspect, body **802** may contain an electrode **816** attached to the outer surface of the container. This electrode **816** may be electrically connected to the return electrode of the stimulation device through port **814**. In this embodiment the integrated system would reduce the need for connection of an additional return current electrode such as **1102**.

[0101] In another aspect shown in FIG. **11**, body **802** may include an attachment device, such as a strap **124** that may facilitate attaching the container **800** to an object, such as a patient's limb, a belt, hospital equipment or the like. Various other attachment devices may be utilized, such as hook and loop materials, magnets, elastic, or the like.

[0102] FIGS. **12-14** illustrate a splitter device **1200** that may operatively split a stimulation signal and/or return signal. According to embodiments, the stimulation device **1200** generally include a housing **1202** that houses circuitry **1220**. The housing may include an input port **1206** and one or more output ports **1212**, **1214**, **1216**, and **1218**. It is noted that splitter device **1200** may include any number of input or output ports. Internal circuitry therein may be utilized to deliver one input signal to multiple output channels, maintain current level at set output, such as for example at 1.0 or 2.0 mA.

[0103] Splitter device **1200** may receive a first wire **1230** via input port **1206**. In an aspect, the first wire **1230** may be coupled to a stimulation device (not shown) that may operatively apply a stimulation signal to the input port **1206**. The circuitry **1220** may split the stimulation signal to one or more of the output port(s) **1212**, **1214**, **1216**, **1218**, which may be coupled to one or more operative elements, percutaneous leads (e.g., percutaneous leads **300**, **400**, etc.), non-percutaneous leads, or connectors as described herein. In an example, a surgeon may place percutaneous leads **400** in different target tissue in one or more patients. The surgeon may attach the percutaneous leads **400** to the splitter device **1200** and may attach a stimulation device to the splitter device. The stimulation device may apply a prolonged stimulation signal to multiple tissue regions based via the splitter device **1200** and/or percutaneous leads **400**.

[0104] As described herein, the circuitry **1220** may be an interface between a stimulation signal (received at input port **1206**) and divides the input signal to one or more output signals with attached electrodes, selectively output at one or more of output port(s) **1212**, **1214**, **1216**, and **1218**. This may allow the stimulation device to utilize additional channels or contacts than otherwise available.

[0105] It is noted that the circuitry **1220** may include a power source (e.g., battery) or may receive power from the stimulation device or other external source. In an example, the circuitry may comprise an inductive circuit that operatively stores power in one or more capacitors. According to another embodiment, the input signal received at input port **1206** may supply the power. The circuitry may include the components to ensure output remains within a specified range (e.g. 2.0 mA) whenever a stimulation pulse is delivered to the input port. Moreover, the circuitry **1220** may include a demodulator (e.g. to receive and decode information) and one or more switches or registers that control output to percutaneous leads **400** or other electrical contacts, or the like, In an aspect, the switches may be utilized to detect whether an electrical contact is connected to output port(s) **1212**, **1214**, **1216**, and **1218** such that power may be selectively applied to the ports.

[0106] In embodiments the splitter device **1200**, or aspects thereof, may be disposed on or within a container (e.g., container **800**), or placed on the body outside the container. For instance, the

housing **1202** may be disposed within layers of material in the container **800**. In another aspect, the circuitry **1220** may be disposed within the material of the container **800** (e.g., the housing may comprise the material of the container). The splitter device **1200** may be sterilized.

[0107] According to various embodiments, splitter device **1200** may comprise a clip, fastener, magnet, etc., that operatively attach the splitter device to a patient, clothing, hospital equipment, or the like.

[0108] FIGS. **15-17** illustrate a stimulation system **1500** primarily comprising a stimulation device **1501**. The stimulation device **1501** may operatively control stimulation of tissue via one or more stimulating medical devices including, for example, stimulation device **1501**, probe **1510**, percutaneous electrodes **300/400**, cutting devices, drilling devices, augers, fixation devices or the like. The stimulation system **1500** may comprise an external stimulator that is selectively attachable ex-vivo a patient. The stimulation system **1500** may comprise a box that may be attached to a patient, such as by adhering to a patch assembly or other adhesive device attached to a patient, selectively attaching the box to clothing of a patient, including a strap or necklace that a patient can wear to hold the box, using Velcro to attach the box to the patient or patient's clothes, and the like.

[0109] In an aspect, the stimulation device **1501** may comprise circuitry that operatively generates an electrical stimulation signal to be applied to a tissue. In an aspect, the stimulation device **1501** may comprise similar functionality as described with reference to other stimulation devices (e.g., stimulation device **102**). Moreover, stimulation device **1501** may comprise a user interface **1508**, including a display **1502**. As described herein, the user interface **1508** may comprise input/output devices that operatively receive user input. A user may interact with the user interface **1508** to adjust parameters of the stimulation signal and/or receive information from the display **1502**. For example, the display **1502** may indicate to a user the length of time of the current therapy has left, the time the therapy has been applied, the number of doses of therapy applied and/or number of therapies yet to be applied. By way of a non-limiting example, if the therapy has an intended duration of an hour and twenty minutes of therapy has been applied, the display **1502** may show that there are forty minutes left on the therapy. The display **1502** may comprise an LED screen that can provide any of the information indicated in the present disclosure to the user, patient and/or clinician. Moreover, stimulation device **1501** may comprise a user interface **1508**, including a display **1502** and audio feedback. In at least some embodiments, the user interface **1508** may generate an alert that is audible, visual, tactile (e.g., vibration), or combination of the above. For instance, the user interface **1508** may generate alerts to indicate that therapy is complete, a lead has moved, a lead has become disengage, stimulation has been disrupted, an error has occurred, a warning (e.g., power supply is low), or the like. As described herein, the user interface **1508** may comprise input/output devices that operatively receive user input. The display **1502** with or without additional audio tones may display to the user signals recorded on the one or more input ports **1506**, which may be connected to other sensors or systems to record physiologic signals such as pressure, electromyograms, stretch, force, or the like. In an aspect, the stimulation device **1501** may include a communication component that may be wired or wireless. For example, the stimulation device **1501** may include a wireless transmitter/receiver configured to communicate via one or more communication protocols (e.g., Wi-Fi, BLUETOOTH, NFC, etc.).

[0110] Stimulation device **1501** may include one or more output ports **1512** that may be operatively coupled with an operative element (e.g., a probe **100**), one or more percutaneous leads **400** and/or connectors **500** (e.g., as shown in FIG. **18**), or other components. According to one or more embodiments, the stimulation device **1501** may include one or more output ports **1506** that may be coupled to a return electrode **1802** via a wire **1804**. It is noted that the return electrode **1802** may comprise a pad-type, needle, or other style electrode as described herein. In an embodiment **1700**, the return electrode **1516** may be affixed to the back of the stimulation device **1512**.

[0111] In embodiments, a first side **1510** of the stimulation device **1501** may comprise user interfaces **1508** as described herein. A second side **1512**, which may be generally opposed to the

first side **1510**, may include return current electrode **1516**. The second side **1512** or sides of the device may operatively receive belt **1530**. Belt **1530** may be coupled to an object, such as a patient, hospital equipment, or the like. It is noted that stimulation device **1501** may be attachable to objects via other components such as clips, magnets, VELCRO, adhesives, or the like. It is further noted that stimulation device **1501** may be disposed within a container (e.g., container **800**), as described with reference to FIGS. **8-11**.

[0112] As described herein, stimulation devices (e.g., stimulation device **102**, **1501**, etc.) may operatively apply prolonged stimulation to target tissue. In examples, the prolonged stimulation may be applied post-op (e.g., after completion of a procedure). For instance, a surgeon may create an incision in a patient. The surgeon may utilize a stimulation device to apply a stimulation signal within the incision. Although in some embodiments, no stimulation signal may be utilized during the surgery. Instead, the stimulation may be applied only after the surgery, such as by way of a non-limiting example, during sensory nerve repair, motor or mixed nerve where clinician does not need to or otherwise utilize a stimulator for nerve identification. It is noted that stimulation may be applied to motor or mixed nerves wherein the surgery does not utilize stimulation during the surgery. Regardless of whether stimulation was applied during surgery or after closing of the incision, the surgeon may place a percutaneous lead in or proximal target tissue. The percutaneous lead may be attached to a stimulation device. The stimulation device may apply a prolonged stimulation. In examples, the stimulation device may be the same stimulation device utilized during a procedure or may be a different stimulation device. In another aspect, the stimulation device may be disposed in a container that generally prevents contamination by or movement of the stimulation device.

[0113] As described herein, after completion of the stimulation, a surgeon may remove the percutaneous lead. In an aspect, the percutaneous lead may be pulled out of target tissue.

[0114] The foregoing is considered as illustrative only of the principles of the invention.

Furthermore, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the invention to the exact construction and operation shown and described. While the preferred embodiment has been described, the details may be changed without departing from the invention, which is defined by the claims.

Claims

1. A method for electrically stimulating tissue, comprising: percutaneously placing a lead within a range of an electrical stimulation signal to a target nerve that is at a site of a nerve injury, wherein the lead is operatively attachable to an electrical stimulation device; applying an electrical stimulation pulse having a waveform directly to the target nerve through the lead and the electrical stimulation device before a surgical procedure; and directly activating excitable tissue of the target nerve for a therapy duration that comprises ten minutes or greater but less than 30 minutes, wherein the direct activation alters recovery of the target nerve.
2. The method of stimulating the tissue of claim 1, wherein the electrical stimulation device is a handheld electrical stimulation device.
3. The method of claim 1, wherein the electrical stimulation device comprises an adapter that is selectively attached to the electrical stimulation device.
4. The method of claim 1, wherein the electrical stimulation pulse comprises a biphasic waveform with controlled current during a cathodic phase.
5. The method of claim 1 further comprising dividing an input signal to a plurality of output signals through a corresponding plurality of electrodes extending from the lead.
6. The method of claim 1, wherein the stimulation device utilizes a plurality of channels to deliver the electrical stimulation.
7. The method of claim 1, wherein the electrical stimulation applied has a frequency of between 10

Hz and 40 Hz.

8. The method of claim 1, wherein the electrical stimulation applied has a frequency of 15 Hz to 20 Hz.

9. The method of claim 1, wherein the electrical stimulation applied has an amplitude between 0.1 mA and 10 mA.

10. The method of claim 1, wherein the electrical stimulation applied has an amplitude of 2 mA.

11. A method comprising: positioning a lead toward a target nerve, wherein the lead is operatively attachable to an electrical stimulation device; applying electrical stimulation to the target nerve through the lead and the electrical stimulation device to determine a location of the target nerve and to assess if the target nerve is injured; placing the lead at or near the injured target nerve; applying an electrical stimulation pulse having a waveform directly to the target nerve through the lead and the electrical stimulation device; and directly activating excitable tissue of the target nerve for a therapy duration that comprises one minute or greater but less than or equal to 10 minutes in order to alter recovery of the target nerve.

12. The method of stimulating the tissue of claim 11, wherein the electrical stimulation device is a handheld electrical stimulation device.

13. The method of claim 11, wherein the electrical stimulation device comprises an adapter that is selectively attached to the electrical stimulation device.

14. The method of claim 11, wherein the electrical stimulation applied has a frequency of between 10 Hz and 100 Hz.

15. The method of claim 11, wherein the electrical stimulation applied has a frequency of between 10 Hz and 40 Hz.

16. The method of claim 15, wherein the electrical stimulation applied has an amplitude between 0.1 mA and 10 mA.

17. The method of claim 11, wherein the electrical stimulation applied has a frequency of 15 Hz to 20 Hz.

18. The method of claim 17, wherein the electrical stimulation applied has an amplitude between 1 mA and 3 mA.

19. The method of claim 11, wherein applying electrical stimulation to the target nerve through the lead and the electrical stimulation device to determine the location of the target nerve and to assess if the target nerve is injured occurs before surgery.

20. The method of claim 11, wherein applying electrical stimulation to the target nerve through the lead and the electrical stimulation device to determine the location of the target nerve and to assess if the target nerve is injured occurs during surgery.

21. A method for electrically stimulating tissue, comprising: placing a lead at or near a target nerve, wherein the lead is attached to an electrical stimulation device; applying electrical stimulation directly to the target nerve through the lead and the electrical stimulation device; and directly activating excitable tissue of the target nerve for a therapy duration that comprises 10 minutes or greater but less than 30 minutes in order to alter recovery of the target nerve, wherein the electrical stimulation applied has a frequency of between 10 Hz and 40 Hz, an amplitude between 1 mA and 3 mA and a pulse duration of 75 microseconds to 125 microseconds.

22. The method of claim 21, wherein application of the electrical stimulation to the target nerve occurs before surgery.

23. The method of claim 21, wherein application of the electrical stimulation to the target nerve occurs during surgery.

24. The method of claim 21, wherein application of the electrical stimulation 2 to the target nerve occurs after surgery.

25. The method of claim 21, wherein placing the lead comprises placing the lead percutaneously.

26. The method of claim 21, wherein the electrical stimulation device is a handheld electrical stimulation device.

